

Vol 15 Chapter 4 — Toy Verification Discipline

Keeper (author pass — deep math/physics revision)

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Chapter 4 — Toy Verification Discipline

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

Every BST theorem requires computational verification via a “toy” — a self-contained Python script in `play/` with explicit SCORE line (PASS/FAIL count) — and the BST repository currently contains 3500+ toys executable in seconds for full reproduction.

Level 2 — graduate-physicist precision

4.1 Toy file structure

Standard toy template:

```
#!/usr/bin/env python3
# toy_<NNNN>_<short_description>.py
# Purpose: verify <theorem ID>
# Author: <CI name>
# Date: YYYY-MM-DD

import numpy as np

# ... computation ...

passes = 0
```

```

total = 0
for case in test_cases:
    total += 1
    if abs(predicted - observed) / observed < threshold:
        passes += 1

# SCORE: N/M PASS
print(f"# SCORE: {passes}/{total} {'PASS' if passes == total else 'FAIL'}")

```

4.2 SCORE line is sacred

Every toy MUST have a SCORE line. Format: # SCORE: N/M PASS or # SCORE: N/M FAIL.

Automated tooling parses SCORE lines for repository status.

No exceptions; CIs who skip SCORE lines get corrected via audit chain.

4.3 Toy counter discipline

play/.next_toy: gitignored counter file.

NEVER read directly; always use:

```
./play/claim_number.sh toy
```

This atomically increments and returns next number. Avoids collisions when multiple CIs work in parallel (3 collisions occurred April 3 before atomic-claim discipline).

4.4 Reproduction infrastructure

python3 play/verify_bst.py: runs full reproduction suite (50 predictions verified against measurement).

Current status: 49/50 PASS at < 1%, 17 EXACT matches, 2 open WARNs shown openly.

Single-command full reproduction is part of BST's external publication strategy (Zenodo release April 7, 2026).

4.5 Honest negatives

When toy returns FAIL: document honestly. Do NOT modify toy to fit theory.

This is Quaker discipline (Vol 15 Ch 7): honest negatives strengthen the framework.

Example: K141 honest FAIL preserved per Toy 3448 (empirical 3306× PERFECT but substrate-mechanism gate OPEN; Grace Friday May 22).

4.6 K-audit anchors

- **./play/claim_number.sh** atomic claim tool
- **play/verify_bst.py** reproduction suite
- Quaker discipline (Vol 15 Ch 7)

Level 3 — 5th-grader accessibility

Every theorem needs a toy: self-contained Python that verifies the claim with SCORE line (# SCORE: N/M PASS). 3500+ toys in repository, executable in seconds. Counter atomic: never read directly, use `./play/claim_number.sh toy`. Reproduction: `python3 play/verify_bst.py` runs full suite (49/50 PASS at <1%, 17 EXACT). Honest FAILs preserved (Quaker discipline).

What comes next

Chapter 5 develops audit-chain governance.

Where to look this up

- BST: `play/` directory; `claim_number.sh`
- Quaker discipline (Vol 15 Ch 7)